

CodeAlpha Internship Report

Task 2: Exploratory Data Analysis (EDA)

Organization: CodeAlpha

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1. Project Overview

This project performs Exploratory Data Analysis (EDA) on a Spotify tracks dataset to understand its structure, assess data quality, identify patterns, and explore relationships among audio features. The analysis provides a strong foundation for visualization and predictive analytics.

2. Objectives

- Understand the dataset structure.
- Clean and preprocess data.
- Identify missing values and duplicates.
- Generate descriptive statistics.
- Detect outliers.
- Analyze correlations between features.

3. Tools and Technologies

Python 3.12+

Libraries: Pandas, NumPy, Matplotlib, Seaborn

Environment: Google Colab / Jupyter Notebook

4. Dataset Information

The Spotify Tracks dataset contains audio features such as popularity, danceability, energy, loudness, acousticness, speechiness, valence, tempo, instrumentalness, liveness, duration, and artist information.

5. Exploratory Data Analysis

Data Inspection: Verified dataset dimensions, data types, missing values, and duplicates.

Statistical Summary: Generated descriptive statistics to understand feature distributions.

Missing Value Analysis: Examined null values and applied appropriate preprocessing where required.

Outlier Detection: Used boxplots to identify unusual observations while preserving meaningful values.

Correlation Analysis: Explored relationships among numerical features using a correlation matrix.

6. Key Findings

Most tracks exhibit medium-to-high popularity. Energy and loudness show a positive correlation, while acousticness generally has an inverse relationship with energy. The dataset contains minimal missing values and valid outliers representing natural variation in music characteristics.

7. Conclusion

The EDA process provided valuable insights into the Spotify dataset, improved understanding of feature relationships, and established a reliable basis for subsequent visualization and machine learning tasks.

Internship Submission Details

Repository: CodeAlpha_DataAnalytics

Task: Task 2 – Exploratory Data Analysis (EDA)

Status: Completed